0417. Pacific Atlantic Water Flow

ium | Category: Graphs | LeetCode: https://leetcode.com/problems/pacific-atlant

Problem

- Water flows to a neighbour with height <= current. The Pacific touches the top and left edges, the Atlantic the bottom and right. Return cells that can reach both oceans.

Intuition

- Searching from every cell is expensive. Reverse the flow: start from each ocean's border and climb to neighbours with height >= current.
- Cells reached by both reverse searches are the answer.

Brute Force

- DFS from every cell to see whether it reaches each ocean: O((mn)^2).

Optimal Approach

- Pac/atl boolean grids.
- DFS from all top/left cells into pac; from all bottom/right cells into atl, moving only uphill (>=).
- Collect cells with pac && atl.

Worked Example

```
5x5 example from LeetCode.

Pacific reverse-DFS starts on row 0 and column 0
Atlantic reverse-DFS starts on row 4 and column 4
intersection: [0,4] [1,3] [1,4] [2,2] [3,0] [3,1] [4,0]
```

Java Solution

```java
public class PacificAtlanticWaterFlow {
    public List<List<Integer>> pacificAtlantic(int[][] h) {
        int m = h.length, n = h[0].length;
        boolean[][] pac = new boolean[m][n], atl = new boolean[m][n];
        for (int r = 0; r < m; r++) { dfs(h, r, 0, pac, h[r][0]); dfs(h, r, n - 1, atl, h[r][n - 1]); }
        for (int c = 0; c < n; c++) { dfs(h, 0, c, pac, h[0][c]); dfs(h, m - 1, c, atl, h[m - 1][c]); }
        List<List<Integer>> res = new ArrayList<>();
        for (int r = 0; r < m; r++)
            for (int c = 0; c < n; c++) if (pac[r][c] && atl[r][c]) res.add(List.of(r, c));
        return res;
    }

    private void dfs(int[][] h, int r, int c, boolean[][] seen, int prev) {
        if (r < 0 || c < 0 || r >= h.length || c >= h[0].length || seen[r][c] || h[r][c] < prev) return;
        seen[r][c] = true;
        dfs(h, r + 1, c, seen, h[r][c]); dfs(h, r - 1, c, seen, h[r][c]);
        dfs(h, r, c + 1, seen, h[r][c]); dfs(h, r, c - 1, seen, h[r][c]);
    }
}
```

Complexity

- Time: O(m n). Space: O(m n).

Pitfalls

- The comparison is reversed (>=) because we walk against the water.